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Module Framework

Statistics and Data Science 14223-188

2026

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1 Introduction

1.1 The purpose of the module framework

This module framework contains important information regarding the content, assessment and administration of the module. The module framework is a document that is regarded as an agreement between the lecturer and student that could be amended during the year. Please take note of any formal announcements on the learning platform EMSLearn in this regard.

1.2 What are statistics and data science?

As a result of the information explosion, decisions are becoming increasingly based on large volumes of data. Statistics is a way to get information from data. Statistics is concerned with the drawing of objective conclusions from investigations where outcomes are subject to uncertainty and variability. Internationally, Statistics is a compulsory service module in almost every study program leading towards a professional qualification. People in this field are appointed amongst others as statisticians, data-mining experts, data scientists, data engineers, and statistical analysts in the marketing, information, and management divisions of companies.

Statistics is also helpful in predicting diseases, medical studies, genetic research, climate change research, political campaigns, insurance investigations, manufacturing consumer goods, performing quality testing and analysing the stock market. Statistics is the process of collecting, measuring, analysing, and communicating information to aid decision making within an organisation. Students trained in Statistics develop the ability for abstract reasoning, enabling them to make informed decisions based on statistical data.

Data Science is the combination of Statistics, Computer Science, and domain knowledge to build data-driven products that help companies extract meaningful information from their data. Data scientists use data to build products for businesses that aid decision making. Statistics and Data Science 188 introduces the fundamental concepts of data science, and present introductory statistical tools used in the data science profession.

1.3 Module aim

Statistics and Data Science 188 is an introductory course in practical data description and data analysis. The objective of the module is to expose students to basic statistical concepts, exploratory data analysis, probability theory, hypothesis testing and regression analysis. All these topics will be applied to datasets using the Excel computer software.

1.4 Module outcomes

After the successful completion of this module you will be able to:

- Define, solve, analyse, and interpret a linear programming problem (online topic for B-Accounting - SAICA requirement).
- Understand basic statistical concepts and definitions.

- Define the fundamental concepts of data science and describe the data cycle.
- Recognise and discuss ethical issues inherent to data science and statistics.
- Effectively create graphical displays of statistical information in datasets.
- Define the basic concepts of probability theory.
- Contextually identify and analyse various probability and sampling distributions.
- Demonstrate fundamental knowledge of statistical inference.
- Statistically model the linear relationship between variables.
- Report on the results obtained from statistical analyses.
- Apply statistical methods to datasets using Excel.

1.5 Language of instruction

This course subscribes to the language option 7.1.4 as described in the Faculty's language plan. In brief, lectures will be offered only in English. Support will be provided as required for Afrikaans-speaking students.

Students are divided into groups based on their particular timetables.

Slides, tutorials and assessments are made available in both English and Afrikaans.

2 Administration

2.1 Communication

The Announcements on EMSLearn serve as a unilateral source of information between lecturers and students. Announcements are formal and binding. The onus is on students to stay up to date with these announcements.

We strongly encourage the use of the **Forum for General Use** on EMSLearn as a primary platform to post any queries or concerns. This platform can become a very useful resource since there is a high likelihood that your query or concern is shared by other students. We also encourage students to assist one another by replying to posts directly. The staff involved on the course will also be keeping an eye on the forum and they will also take turns to respond.

You can discuss personal concerns or make an appointment with your lecturer directly. Their email addresses are indicated below. Should the **Forum for General Use** and/or the course email or queries raised with your lecturer not be sufficient, you can contact the course convenor, Ms Zoë-Mae Adams. Please do not contact the convenor before exhausting the other alternatives.

Should all the above fail to address your concerns, an appeal can be directed to the chairman of the Department, Professor Justin Harvey at jharvey@sun.ac.za. The department telephone number is (021) 808 3244 and the physical address is: 2nd Floor – Entrance 5, Van der Sterr building.

2.2 Module convenor

Name	Office	E-mail address	Group
Ms Z Adams	Van der Sterr 2003	zadams@sun.ac.za	Group 3

2.3 Lecturer & additional information

Name	Office	E-mail address	Group
Dr MMC Lamont	Van der Sterr 2022	mmcl@sun.ac.za	1st semester Group 1
Mr H Sadiq	Van der Sterr 2028	hassans@sun.ac.za	2nd semester Group 2
Mr H Bakker	Van der Sterr 2007	hbakker@sun.ac.za	Group 5
Ms K Stapelberg	Van der Sterr 2009	kstapelberg@sun.ac.za	Groups 4 and 6
Mr C Hildebrand	Van der Sterr 2009	24821950@sun.ac.za	1st semester Group 2, 2nd semester Group 1

Please note: Appointments with lecturers need to be arranged directly with the lecturers through their email addresses.

2.4 Groups

At registration students will automatically be placed into their designated groups.

Students are divided into groups for lectures and practical sessions and will mostly be allocated to different lecture and practical groups.

Please note: a student may for example be allocated to group 2 for lectures and group 6 for their practical sessions.

It is important that students remain in their allocated groups. Failure to do so can lead to overcrowding.

2.5 Lecture times

Consult MySun for details of your groups' lecture and practical session times and venues. The attendance of all activities are compulsory.

3 Resources

3.1 Prescribed textbook

Statistics for Managers Using Microsoft Excel, 9th Edition

Authors: Levine, Stephan, Krehbiel, and Szabat.

Printers: Pearson

3.2 Additional resources

- **Computer software: Microsoft Excel**

  The course makes extensive use of Microsoft Excel to demonstrate concepts related to statistics and data science. Students have access to computers with the required software during the practical lectures. Students are also allowed to make use of the computer rooms on campus to practice using Excel. Please note that Microsoft Office 365 (Word, Excel, PowerPoint) is freely available, and students can download it via MySUN.

- **Calculator (compulsory): HP 10bII+**

  A document on how to use the calculator is available on EMSLearn. Note this calculator is also compulsory for Theory of Interest 152 and has additional statistical methods that standard scientific calculators do not.

4 Learning methodology

4.1 Method of instruction

Principles of adult learning, with an emphasis on blended learning, will be applied. You are invited to participate interactively in this module and ask questions whenever anything is unclear.

There are three formal lecture periods and there is one practical session, consisting of two periods, every week for each group.

Lectures will take place in lecture halls and will be used to explain the theory and applications of the different topics in Statistics and Data Science. The practical sessions will take place in a computer venue where lecturers will demonstrate the application of statistical concepts and the use of Excel.

Students are strongly encouraged to attempt the tutorial questions on their own before being shown how to do them.

4.2 Practical sessions

Tutorials on every chapter will be made available for students to work on in their own time. The weekly practical sessions will be used by lecturers to illustrate the methods used in solving the tutorial problems. Attendance at all practical and lecture sessions are compulsory.

Memoranda in both English and Afrikaans for the tutorials will be made available after the relevant practical sessions.

It is important to attempt the problems yourself before the answers are discussed in the practical sessions.

4.3 EMSLearn

EMSLearn is an integral part of this module and should be consulted daily for announcements, resources, communication and learning activities. EMSLearn is the platform that is used to electronically link students and lecturers and where, for example, messages can be exchanged, questions can be asked and extra information about courses can be obtained. As a registered student for Statistics and Data Science 188 you automatically have access to the module on EMSLearn. When you log in to EMSLearn, Statistics and Data Science 14223-188 will be one of the modules you can choose from.

4.4 Linear Programming for B.Acc Students

Students in B.Accounting are required to complete a course in Linear Programming. The online resource will be made available at the start of the midyear recess. During the first week of the third term B.Acc students are required to complete a quiz assessment on EMSLearn. These results will be passed on to the School of Accounting as part of the requirements of the South African Institute of Chartered Accountants and will be considered as part of the final mark calculations.

Please note, only students registered in a B.Accounting are required to do this. For example, students in a B.Com Management Accounting or a B.Com with accounting subjects are not required to complete this.

4.5 Copyright

Stellenbosch University holds the copyright to all material in this module. Material which is made available through EMSLearn may be subject to certain licencing conditions. You may only use the material for your own private study and research at the University and not for any other purpose. Printed material is also only made available for class discussion, private study and research purposes. Further reproduction or distribution of such material is prohibited. Contravention of these rules will result in disciplinary action and may result in legal action.

5 Assessment

The Faculty's Assessment Rules are available online at `www.su.ac.za/en/faculties/economy/current-students`. Please watch the information video on your module's EMSLearn page and make sure you read the section on Assessment in the module framework carefully. The faculty's approach to assessment places the responsibility of learning with the student. Therefore, you must make use of all available assessment opportunities as far as possible and ensure you understand how assessment in the module works.

5.1 Main assessments

This course has two main assessments per semester. The first assessment in each semester (A1S1 and A1S2) will be a computer-based assessment primarily focused on testing the student's competency in Excel. The second main assessment in a semester (A2S1 and A2S2) will be written. Further details will be made available during the semesters.

5.2 Assessment Further (AF)

The objective of the AF is to promote continuous learning throughout the module. It gives students the opportunity to regularly test their knowledge, and quickly identify areas that require more work.

These take the form of regular online quizzes that students will have one week to complete. These assessments AFS1 after the first semester and AFS2 in the second will each contribute 5% to the final mark. For each semester, the lowest mark will be discounted in the calculations.

Please refer to EMSLearn for the dates when each online assessment is available.

Each AF quiz will be available for one week and will cover the content of a specific chapter. Students get one attempt during the week for which they have one hour to complete the assessment.

Please note: B.Accounting students have an additional Linear Programming AF, which will form part of their AFS2 mark. Note, only the lowest non-Linear Programming mark will be dropped in the calculations.

5.3 Third assessment (A3)

For students who have missed a main assessment and students who have not achieved a pass mark at the end of the year, a final opportunity to pass is provided in the form of a second exam (A3). Details of access to the A3 are provided below.

5.4 Assessment opportunities

Assessment 1 Semester 1 (A1S1) Thursday 23 April Details to be confirmed Weight: 0.10
Assessment 2 Semester 1 (A2S1) Monday 01 June 09h00 Weight: 0.25
Assessment Further Semester 1 (AFS1) Weight: 0.05
Assessment 1 Semester 2 (A1S2) Saturday 29 August Details to be confirmed Weight: 0.25
Assessment 2 Semester 2 (A2S2) Monday 09 November 09h00 Weight: 0.3
Assessment Further Semester 2 (AFS2) Weight: 0.05
Assessment 3 (A3) Tuesday 01 December 14h00 Weight: 0.4

5.5 Religious and cultural days

In addition to public holidays, during which the University will be closed for business, no assessments will be scheduled on specific religious and cultural days that are noted in the SU Almanac. If any main assessment is scheduled on a religious or cultural day not included in the almanac, it is the student's responsibility to request special arrangements. The following apply:

- For A2S1, A2S2 and A3 the Assessment Office will make the necessary arrangements where possible. Contact the Assessment Office in the course of the last two class weeks of the semester.
- For A1S1 and A1S2 the department/school will make the necessary arrangements if it is feasible in accordance with the arrangements usually made by the Assessment Office for A2S1, A2S2 and A3. If a student wants to request special arrangements, they must contact the lecturer at least two weeks before **each** assessment opportunity.

5.6 Calculation of marks

[Please note that Example calculations in this section does not cover all possible scenarios]

The calculation of your final mark (FM) is based on a weighted formula. The formulas are as follows:

MY (Mid-Year) mark calculated after A2S1:

$$MY = (\frac{w_{\text{AFS1}}}{w_{\text{sum}}})\text{AFS1} + (\frac{w_{\text{A1S1}}}{w_{\text{sum}}})\text{A1S1} + (\frac{w_{\text{A2S1}}}{w_{\text{sum}}})\text{A2S1}$$

where w_{sum} is the sum of the weights for the assessments a student has completed.

Note, the weight of the AFS1 will be included irrespective of whether a student wrote them.

Students will receive a MY mark only if they wrote at least one of A1S1 and A2S1.

MTD (Mark To Date) calculated after A1S2 but before A2S2:

$$MTD = (\frac{w_{\text{AFS1}}}{w_{\text{MTD}}})\text{AFS1} + (\frac{w_{\text{A1S1}}}{w_{\text{MTD}}})\text{A1S1} + (\frac{w_{\text{A2S1}}}{w_{\text{MTD}}})\text{A2S1} + (\frac{w_{\text{AFS2}}}{w_{\text{MTD}}})\text{AFS2} + (\frac{w_{\text{A1S2}}}{w_{\text{MTD}}})\text{A1S2}$$

where w_{MTD} is the sum of the weights for the assessments a student has completed.

Note, the weight of the AFS1 and AFS2 will be included irrespective of whether a student wrote them.

Example calculation 1 (completed all assessments):

Marks obtained:

- AFS1: $60/100$
- A1S1: $75/100$
- A2S1: $40/100$
- AFS2: $60/100$
- A1S2: $55/100$

$w_{\text{MTD}} = 0.05 + 0.1 + 0.25 + 0.05 + 0.25 = 0.7$

$$\begin{aligned} MTD &= (\frac{0.05}{0.7}) * 60 + (\frac{0.1}{0.7}) * 75 + (\frac{0.25}{0.7}) * 40 + (\frac{0.05}{0.7}) * 60 + (\frac{0.25}{0.7}) * 55 \\ &= 4.29 + 10.71 + 14.29 + 4.29 + 19.64 \\ &= 53.21 \end{aligned}$$

Example calculation 2 (missed one assessment):

Marks obtained:

- AFS1: $60/100$
- A1S1: $75/100$
- A2S1: $40/100$
- AFS2: $60/100$
- A1S2: missed

$w_{\text{MTD}} = 0.05 + 0.1 + 0.25 + 0.05 + 0 = 0.45$

$$\begin{aligned} MTD &= (\frac{0.05}{0.45}) * 60 + (\frac{0.1}{0.45}) * 75 + (\frac{0.25}{0.45}) * 40 + (\frac{0.05}{0.45}) * 60 + (\frac{0}{0.45}) * 0 \\ &= 6.67 + 16.67 + 22.22 + 6.67 + 0 \\ &= 52.22 \end{aligned}$$

Example calculation 3 (missed two assessments):

Marks obtained:

- AFS1: $60/100$
- A1S1: missed
- A2S1: $40/100$
- AFS2: $60/100$
- A1S2: missed

$w_{\text{MTD}} = 0.05 + 0 + 0.25 + 0.05 + 0 = 0.35$

$$\begin{aligned} MTD &= (\frac{0.05}{0.35}) * 60 + (\frac{0.1}{0.35}) * 0 + (\frac{0.25}{0.35}) * 40 + (\frac{0.05}{0.35}) * 60 + (\frac{0.25}{0.35}) * 0 \\ &= 8.57 + 0 + 28.57 + 8.57 + 0 \\ &= 45.71 \end{aligned}$$

FM1 (First Final Mark) calculated after A2S2:

$$FM1 = (\frac{w_{\text{MTD}}}{w_{\text{sum}}})\text{MTD} + (\frac{w_{\text{A2S2}}}{w_{\text{sum}}})\text{A2S2}$$

where $w_{\text{sum}} = w_{\text{MTD}} + w_{\text{A2S2}}$.

FM1 is assigned only if you wrote at least two of the following three assessments, namely A1S1, A2S1, A1S2, in addition to having written A2S2.

Example calculation 4 (completed all assessments):

Marks obtained:

- AFS1: $60/100$
- A1S1: $75/100$
- A2S1: $40/100$
- AFS2: $60/100$
- A1S2: $55/100$
- A2S2: $65/100$

$w_{\text{sum}} = 0.7 + 0.3 = 1$

$$\begin{aligned} MTD &= 53.21 \\ FM1 &= (\frac{0.7}{1}) * 53.21 + (\frac{0.3}{1}) * 65 \\ &= 37.25 + 19.50 \\ &= 56.75 \end{aligned}$$

Example calculation 5 (missed one assessment):
Marks obtained:

- AFS1: $60/100$
- A1S1: $75/100$
- A2S1: $40/100$
- AFS2: $60/100$
- A1S2: missed
- A2S2: $65/100$

$w_{\text{sum}} = 0.45 + 0.3 = 0.75 \to 1$

$$\begin{aligned} MTD &= 52.22 \\ FM1 &= (\frac{0.45}{1}) * 52.22 + (\frac{0.3}{1}) * 65 \\ &= 23.50 + 19.50 \\ &= 43 \end{aligned}$$

If a student missed for example A1S1 and A1S2, and still completed A2S2, the FM1 is not assigned.

If a student missed the A2S2, they must complete the A3. (Refer to 5.7 Pass requirements)

FM2 (Second Final Mark) calculated after A3:

$$FM2 = (\frac{w_{\text{MTD}}}{w_{\text{sum}}})\text{MTD} + (\frac{w_{\text{A2S2}}}{w_{\text{sum}}})\text{A2S2} + (\frac{w_{\text{A3}}}{w_{\text{sum}}})\text{A3}$$

where $w_{\text{sum}} = w_{\text{MTD}} + w_{\text{A2S2}} + w_{\text{A3}}$ if you wrote A2S2 and A3, or $w_{\text{sum}} = w_{\text{MTD}} + w_{\text{A3}}$ if you wrote A3 and not A2S2.

FM2 is assigned only if you wrote at least three main assessments of which at least one must be A2S2 or A3.

Example calculation 6 (completed all assessments):

Marks obtained:

- AFS1: $50/100$
- A1S1: $25/100$
- A2S1: $40/100$
- AFS2: $50/100$
- A1S2: $55/100$
- A2S2: $47/100$
- A3: $66/100$

$w_{\text{sum}} = 0.7 + 0.3 + 0.4 = 1.4$

$$
\begin{aligned}
MTD &= 44.64 \\
FM1 &= 45.35 \\
FM2 &= (\frac{0.7}{1.4}) * 44.64 + (\frac{0.3}{1.4}) * 47 + (\frac{0.4}{1.4}) * 66 \\
&= 22.32 + 10.07 + 18.86 \\
&= 51.25
\end{aligned}
$$

Example calculation 7 (missed the A2S2):

Marks obtained:

- AFS1: $50/100$
- A1S1: $25/100$
- A2S1: $40/100$
- AFS2: $50/100$
- A1S2: $55/100$
- A2S2: missed
- A3: $66/100$

$w_{\text{sum}} = 0.7 + 0 + 0.4 = 1.1$

$$
\begin{aligned}
MTD &= 44.64 \\
FM2 &= (\frac{0.7}{1.1}) * 44.64 + (\frac{0}{1.1}) * 0 + (\frac{0.4}{1.1}) * 66 \\
&= 28.41 + 0 + 24.00 \\
&= 52.41
\end{aligned}
$$

When calculating the MY mark, MTD or a FM using the formulas please note the following:

- w_{sum} is the total weight of all the assessment opportunities you completed. Even if you missed main assessment opportunities the total weight will at least be 1.
- For all missed main assessment opportunities. (A1S1, A2S1, A1S2, A2S2 and A3), the weight of the relevant assessment will be reduced to zero in the mark calculation formula (however w_{sum} will never be less than 1).
- Your FM for the module is the highest of FM1 or FM2.

5.7 Pass requirements

A **valid** FM of **at least 50** is required to pass the module. Please take note in the calculation of final marks of the requirement that the **sum of weights needing to be at least one**. This implies a penalty for missed main assessments.

To obtain a **valid final mark** you must:

- Complete at least one of A2S2 or A3
- Complete at least three main assessment opportunities

If you do not obtain a **valid** final mark, you will fail the module.

5.8 Access to Assessment 3 (A3)

Note that it is not necessary to submit medical certificates or any other documents for missed assessment opportunities.

You have access to A3 if either of the following applies:

- If you completed all the main assessment opportunities other than A3 (A1S1, A2S1, A1S2, A2S2) but have not obtained a FM of 50 or more, you have access to A3 in order to improve your FM to 50.
- You missed one or two main assessment opportunities (A1S1, A2S1, A1S2, A2S2) for whatever reason. In this case, a FM of 50 or more can still be obtained if you make use of A3.

No further assessment opportunities will take place after A3.

If you passed the module after A2S2 and you have access to A3 (because you missed a main assessment) you must inform the Registrar's Office through the relevant procedure on SUNStudent of your intention to use the A3 assessment opportunity. This notification must be submitted by no later than 48 hours after the final marks after A2S2 have been released.

Take note: If you are a final-year student and make use of A3, graduation will only be possible in March/April of the next academic year.